

# Area of Regular Polygons

Lesson 5-4

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_ **Class:** \_\_\_\_\_

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## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can find the area of a regular polygon by decomposing it into triangles.

**Language Objective:** I can explain my steps using the words regular polygon, decompose, triangle, and composite.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Regular Polygon

Decompose

Triangle

Composite

Formula



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A polygon with all sides and all angles equal is a .

2

To break a shape into smaller shapes is to  it.

3

A three-sided polygon is a .

4

Made of two or more shapes combined, a figure is .

5

A math rule written with symbols is a .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**You matched each regular polygon to its number of center triangles. What pattern connects the number of SIDES to the number of TRIANGLES, and why?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐

**Regular Polygon** — A shape where all sides and all angles are equal.

*Polígono regular — Una figura donde todos los lados y ángulos son iguales.*

☐

**Decompose** — To break a shape into smaller, simpler shapes.

*Descomponer — Separar una figura en figuras más pequeñas y simples.*

☐

**Triangle** — A shape with three sides.

*Triángulo — Una figura de tres lados.*

☐

**Composite** — Made by putting two or more simple shapes together.

*Compuesto — Formada al juntar dos o más figuras simples.*

☐

**Formula** — A math rule written with symbols.

*Fórmula — Una regla matemática escrita con símbolos.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**I split each polygon into \_\_\_\_ triangles.**

*Dividí cada polígono en \_\_\_\_ triángulos.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** We decompose the hexagon into triangles because we already know how to find a triangle's area.

**Evidence:** Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.

**Reasoning:** This evidence connects the definition of regular polygon to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **I split each polygon into \_\_\_\_ triangles.**

*Dividí cada polígono en \_\_\_\_ triángulos.*

- **The number of triangles equals the number of \_\_\_\_ because \_\_\_\_.**

*El número de triángulos es igual al número de \_\_\_\_ porque \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*